

DTU



25347 Fundamentals of Underwater Acoustic and Optical Systems

ROV Inspection for Harbour Piles

Group 3 – Piles Team 1

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Project Objectives & Research Question

- Problem: The need to evaluate both the structural integrity and the ecological state of underwater piles.
- Research question:
- Generate a high-resolution 3D reconstruction of the piles located in Skoshoved Havn using a multi-camera ROV system.
- Assess biological growth, such as mussels, algae, and other marine organisms attached to the pile surface.
- Estimate the pile diameter with millimeter level precision.

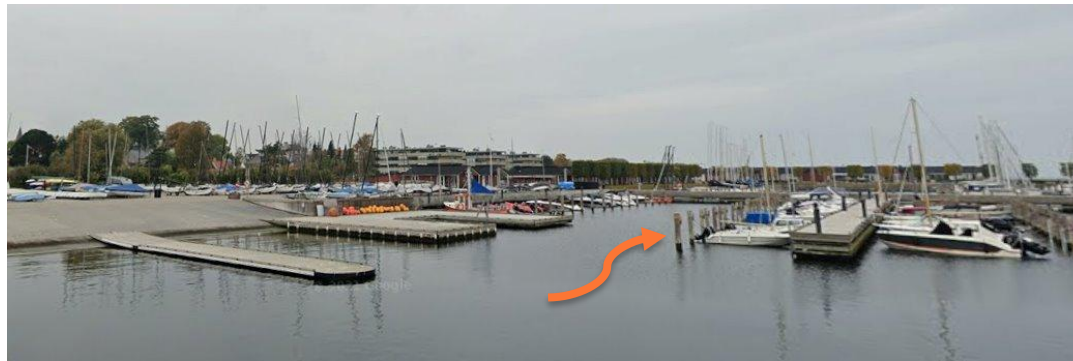


Fig. 1: Google maps street view of the Skovhoved Havn

System & Data Overview

- Inspections conducted on a pile in Skovshoved Havn.
- BlueROV4 equipped with a 4-camera system.
- P2-Piles dataset, containing raw video recordings and calibration parameters, collected on 8 April 2026.
- Pre-computed camera calibration parameters are used as reference for internal camera settings.
- Optical underwater systems are constrained by turbidity and lighting conditions, which can affect the photogrammetry process.
- Processing high-resolution multi-camera data requires computational power, relying on RAM and GPU capabilities.

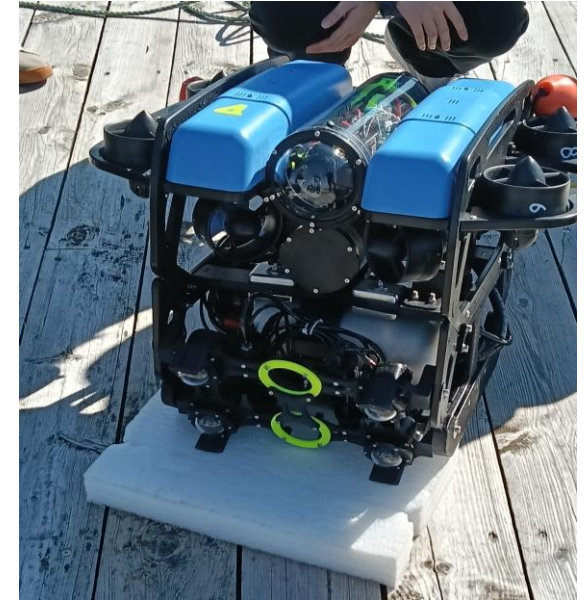


Fig. 2: The multi-camera ROV system

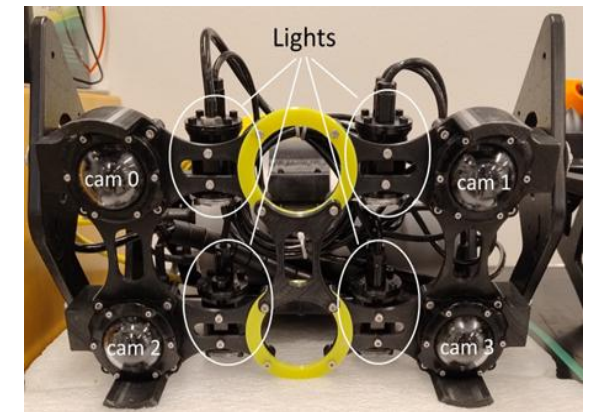
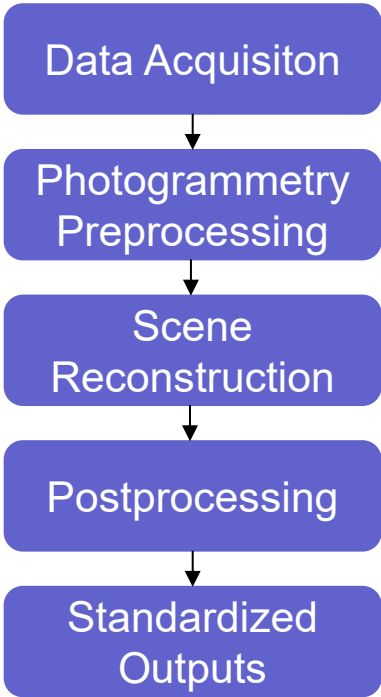


Fig. 3: The multi-camera system (Javier manual)

Theoretical Foundation

- Light behaves differently underwater due to basic optical propagation principles like absorption and scattering.
- The water absorbs certain colours and floating particle scatter the light.
- Light also refracts when it moves from the water into the camera.
- Underwater lights are used to improve the conditions so the images look clear.
- The 3D model accuracy depends on camera geometry and high image resolution instead of acoustic pulse lengths or frequency effects.
- These principles directly limit how well the detection and classification can be performed.

Project Workflow & Time Plan



- Collecting data and performing multi-camera calibration duering data acquisition phase.
- Taking the raw video data from all four cameras and perform phtogrammetry preprocessing to extract keyframes and enhance the images.
- Processed frames go through the scene reconstruction stage where the software builds a sparse point cloud, a dense point cloud, and a three-dimensional mesh.
- Applying postprocessing steps to refine the model orientation and create the final 3D model of the pile.
- The final model allows us to generate standardized outputs such as 3D tiles for further structural and biological analysis.

	April																May					
Task	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	1	2	3	4	5	6
Photogrammetry preprocessing																						
Scene reconstruction																						
Results																						
Presentation																						

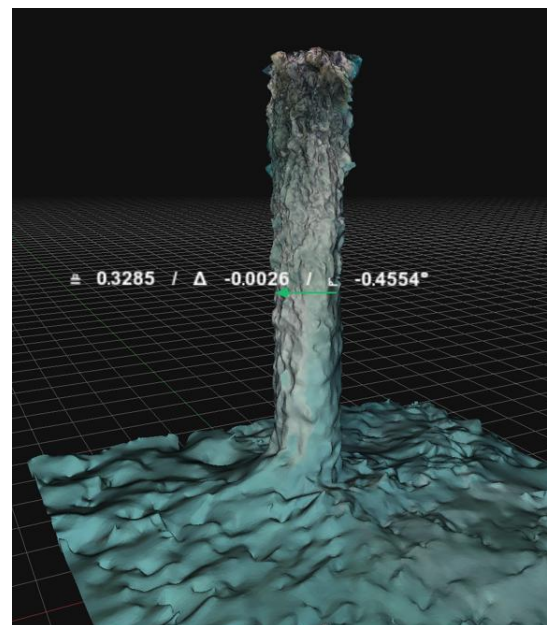
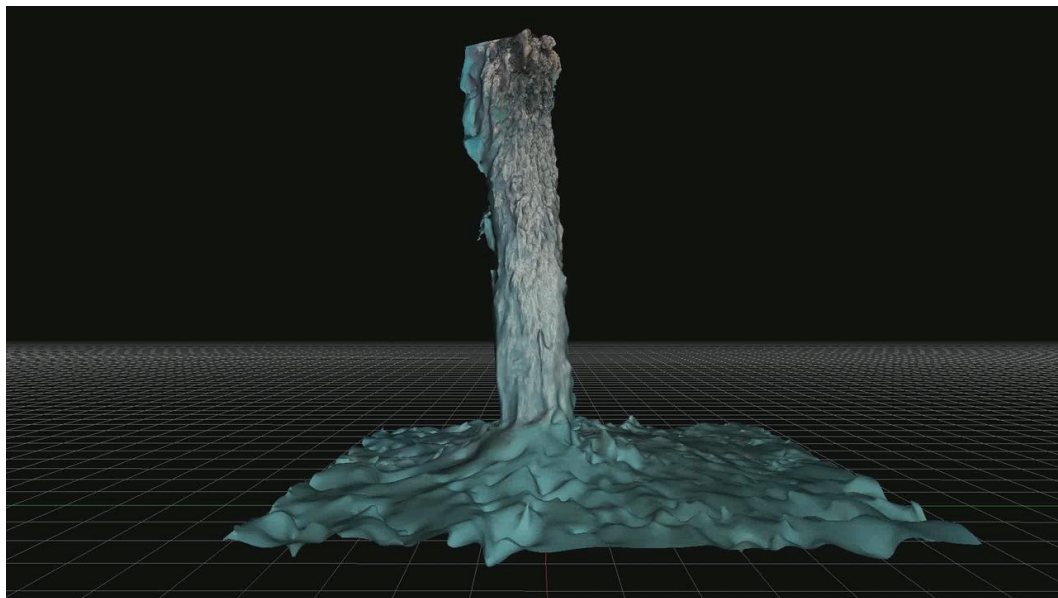
Methods I: Data Processing and Assumptions

- This slide should explain how the data were prepared.
- Examples:
 - Data cleaning
 - Calibration or normalization
 - Coordinate/time matching
 - Filtering
 - Noise handling
 - Selection of transects or test areas
 - Assumptions made

Methods II: Algorithms, Models, or Analysis Approach

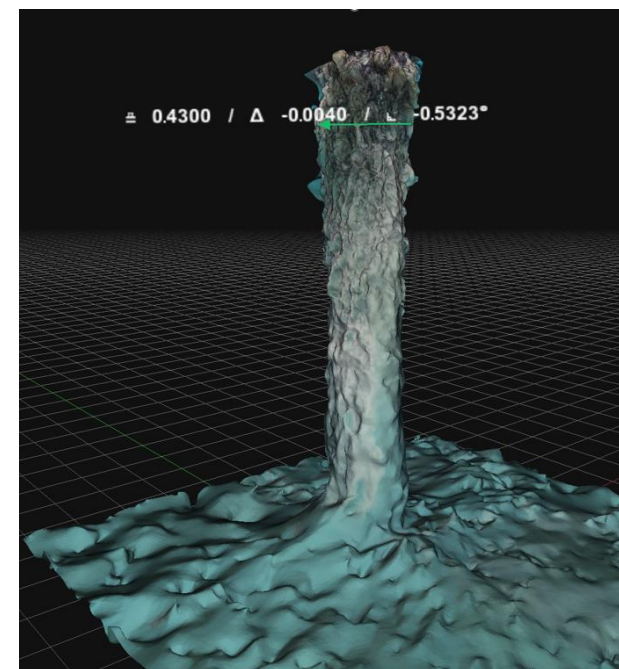
- Describe the actual method.
- Example:
 - Threshold-based detection
 - Peak detection
 - Edge detection
 - Model-based fitting
 - Comparison with reference data
 - Error metrics such as mean error, RMSE, bias, standard deviation
 - For optical projects, this could instead include image segmentation, feature extraction, classification, or visibility estimation, etc, etc
 - etc

Results



Mid-section thinning

Top growth accumulation



Results

- Biological assessment from video

Discussion: What Worked and What Did Not

- Students should interpret their results
- (Examples of questions
 - Which method worked best?
 - Under what conditions?
 - What failed or was unreliable?
 - Were the results physically reasonable?
 - How do the results relate to the theory?)

Conclusion

- A strong conclusion slide has 3–5 concise statements, not a paragraph.
- Add here a reference to the availability of the code, media, other material, etc. (where can we find that)

Improvements & Future Work

Individual Contributions

- The project is primarily assessed as a group outcome. The individual contribution slide is used to document each student's role, level of participation, and specific contributions to the project activities.

Student name	Main contribution	Output	Estimated effort in hours

Appendix

- Add here anything that is important but does not fit in the previous slides
- Please for each of them add individual names contribution following the table used before

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